

Daniel Hardesty Lewis

Causal ML Engineer – forecasting, validation, distributed systems

Causal ML engineer building production forecasting systems, validation pipelines, and distributed infrastructure across geospatial & time-series workloads.

Core Expertise

- 10 years **Causal ML**: Build causal inference, forecasting, and evaluation systems that separate signal from confounders and support decision-making
- 8 years **ML Engineering**: Build and operate AI/ML training and inference pipelines, using rigorous validation protocols to ensure model generalization and reliability
- 8 years **Backend & Data**: Build schemas and ETL; ship API-backed services using Python, SQL, and Postgres/PostGIS
- 8 years **Distributed Systems**: Scale batch workloads across distributed systems, profile bottlenecks, and optimize latency & throughput with explicit cost-performance tradeoffs
- 8 years **Technical Leadership**: Drive cross-functional execution via specifications and governance; mentor engineers through design reviews, code reviews, and instruction

Technologies

Causal AI Modeling	DoWhy; EconML; PyWhy; CausalNex; Double ML; Causal Forests
LLM apps	PyTorch; TensorFlow; scikit-learn; Transformers; LoRA/QLoRA/PEFT
Data stack	embeddings; RAG; LangChain; LangGraph; DSPy; FAISS
Infrastructure	Python; SQL; Pandas; Polars; PyArrow; Parquet; Postgres; PostGIS; Redis; Elasticsearch; Supabase
Additional	Linux/Unix; Bash; Git; Docker; Kubernetes; Modal; GCS; Ray; Spark; Dask; Airflow; CI/CD; observability
	CUDA; C++; TypeScript; React

Experience

Professional

- Jun. 2026 – Present **Entrepreneur-in-Residence**, *South Park Commons*, New York, NY
Selective technology community for founders at a leading venture capital firm.
 - Designed and implemented the next iteration of Homecastr’s machine learning architecture: a full-sequence inpainting editable world model.
 - Conducted customer discovery with potential ICPs, including Parcel, Crusoe, and Cushman & Wakefield, to align predictive capabilities with enterprise requirements.
- Sep. 2025 – Present **Founder**, *Homecastr*, New York, NY
Consumer AI forecasting platform serving **66K+ unique visitors** and **4.3M weekly API requests**.
 - Architected a nationwide parcel-scale diffusion forecasting model trained on 50M+ parcel-year observations, reaching **7.0% one-year error** and **12.6% MdAE at 4-year horizons**.
 - Built an **LLM orchestration layer with 30+ function-calling tools** for geographic and comparable-data retrieval; dual-model routing between Claude and GPT-4o Mini optimized cost and latency.
 - Owned the end-to-end pipeline from data ingestion to real-time serving, and used PIT / coverage diagnostics plus feedback from **a16z** and **MetaProp** to sharpen product positioning.
- Oct. 2024 – Sep. 2025 **Technical Advisor**, *PoliBOM*, Seattle, WA
AI tariff mitigation and trade compliance product for manufacturers, centered on agentic BOM analysis and policy scenario simulation.
 - Engineered an agentic retrieval-augmented pipeline for BOM parsing and tariff simulation, replacing manual trade compliance workflows with predictable multi-step reasoning.
 - Directed development of a **conversational AI interface** enabling decision-makers to instantly **retrieve BOMs and receive proactive compliance alerts**.
 - Executing customer discovery with ScoutyAI and Columbia economists, and secured early market validation by presenting the prototype to KPMG.
- Nov. 2023 – Aug. 2025 **Founder**, *Summit Geospatial*, New York, NY
High-resolution elevation data and web delivery for engineering and hazard workflows.
 - Engineered a **statewide seamless Texas DEM mosaic**, resampling 0.5 m LiDAR sources to 1.2 m via nearest-neighbor across **70+ state, federal, and local elevation datasets**.
 - Developed the web distribution platform end-to-end including **data pipeline, tiling, hosting, delivery UX** to support engineering and planning use cases.

- Aug. 2021 – **Senior Data Scientist & Technical Lead**, [Texas Advanced Computing Center \(TACC\)](#), Austin, TX
- Aug. 2023
- Spun out Summit Geospatial alongside UT's VPs of IP to commercialize state-level terrain models from the \$40M [Texas Disaster Information System \(TDIS\)](#).
 - Developed inter-agency RFP and MOU frameworks with state and federal partners, translating technical requirements into scope and milestones for [Real-Time Inundation Mapping](#) and TDIS.
 - Scaled climate and flood models on supercomputers, executing 800,000-node distributed jobs while managing \$1M+ compute budgets and federal partnerships with NOAA, NSF, USACE, GLO.
 - Developed high-resolution 1m flood maps for Texas Emergency Management and compound flood-hazard models with the US Army Corps of Engineers; contributed research supporting Paola Passalacqua's [2022 EGU Bagnold Medal](#) and reviewed Frontera computing allocations.
- Feb. 2018 – **Data Scientist & Research Engineer**, [Texas Advanced Computing Center \(TACC\)](#), Austin, TX
- Jul. 2021
- Competed in [DARPA World Modelers](#) to improve disaster resiliency and food security modeling in East Africa.
 - Mentored Petrobras engineers in deep learning and HPC to enable institutional technology transfer.
 - Cited by AAAI President Yolanda Gil in her [farewell address](#) for contributions to automated scientific modeling.

Research

- Sep. 2024 – **Graduate Thesis Researcher**, [Columbia University GSAPP](#), New York, NY
- May 2026
- Formulated **S-IRM** and **I-DML** (Invariant DML with ICP pre-filtering) to extract environment-invariant causal features and reduce spatial wealth confounder bias.
 - Designed an **Adversarial Causal Spatial Autoencoder** (PyTorch) using minimax gradient reversal to disentangle spatial opposition geometry from neighborhood wealth.
- May 2025 – **Research assistant to Dir. Electrical Engineering Zoran Kostic**, [Columbia University](#), New York, NY
- Aug. 2025
- Math benchmarking and finetuning of Deepseek's GRPO algorithm
- Jan. 2025 – **Research assistant to Dir. Financial Engineering Ali Hirsra (Industry-sponsored research)**, [Columbia University](#), New York, NY
- Oct. 2025
- Developed a multi-asset CVAE latent factor model with Skew-T Mixture priors; achieved 85% R^2 vs. commercial SaaS (75%) and Fama-French (58%) on backtested holdout.
 - Derived a tractable, axiom-compliant per-feature SHAP algorithm using Hessian-based recursive clustering to provide stable temporal regime-dependent attributions for deep learning models.
- July 2019 – **Summer research intern to Pres. AAAI Yolanda Gil**, [University of Southern California](#), Los Angeles, CA
- Aug. 2019
- Integrated hydrological models into the [MINT Platform](#), competing year-long with MITRE to provide DARPA the "Airflow for science"

Teaching

- Jan. 2018 – **Co-instructor**, [The University of Texas at Austin](#), Austin, TX
- Dec. 2020
- Helped design and teach graduate-level courses: *Machine Learning for the Geosciences* for Petrobras engineers, *Intelligent Systems for the Geosciences*, and *Scientific Computation*.

Selected Publications

- 2026 **Hardesty Lewis, D.**, "Predicting Formal Protest Petitions Against Housing Development: Evidence from Austin, Texas". M.S. Thesis, Columbia University GSAPP, 2026. github.com/dhardestylewis/thesis
- 2021 Gil, Y., et al. (incl. **D. Hardesty Lewis**), "Artificial Intelligence for Modeling Complex Systems: Taming the Complexity of Expert Models to Improve Decision Making". *ACM TüS*, 2021. DOI: [10.1145/3453172](https://doi.org/10.1145/3453172)

Education

- 2026 **M.S., Urban Planning**, [Columbia University](#), New York City, NY
- 2017 **B.S., Pure Mathematics**, [University of Texas](#), Austin, TX

Key Projects

- Causal zoning opposition Developed the first property- and owner-level causal ML model of NIMBYism: Double Machine Learning (DML) over a 193k case-period panel to isolate the spatial tax of formal protest petitions on allowable building height.
- M&A predictor Built M&A prediction model (Oct.–Dec. 2025) with Prof. Eric Talley on 1.5M Compustat firm-quarters; long-only portfolio beat S&P 500 by 2%/yr (2004–2020), peaking at 6%/yr.

Languages

- Spanish Native proficiency; French professional working proficiency